Network layer (Basics)

Now lets come to the 3rd layer — Network layer.

If you remember we did talk about transport layer providing process to process communication, now network layer provides host to host communication.

- Network layer exists in:
  - Routers
  - END devices.

Services & protocols:

→ transport segments from sending to receiving hosts.

Sender: encapsulates segments into datagrams & passes to link layers.

Receiver: Delivers ~~data~~ segments to transport layer protocol.

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Protocol: IP

network layer has one protocol only - IP.

→ In all previous layers we also only had end devices but now we also have Routers. as Intermediate devices.

Routers: • examine header fields in all IP datagrams passing through it.

• Then it moves them from input to correct output ports to transfer datagrams along end-end path.

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2 key to functions:

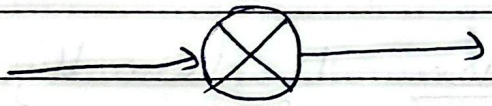
We already know these both:

1) Routing
2) forwarding.

1) Forwarding:

Local per route function

Transferring a packet from a router's incoming link interface to the appropriate outgoing link interface based on the packet's header.



Forwarding:

* Handled in hardware using forwarding table

"DATA PLANE"

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2) Routing:

Network-wide Action

Determining the end-end path that packets take from the source host to the destination host.

~~Routing~~ executed in Software using Routing Algorithms.

"Control Plane"

Data Plane & Control Plane:

Dataplane: Is basically that decides forwarding.

Control Plane: network-wide logic to Routing.

There are 2 control plane approaches:

1) traditional Routing algos:
implement the software in every router.

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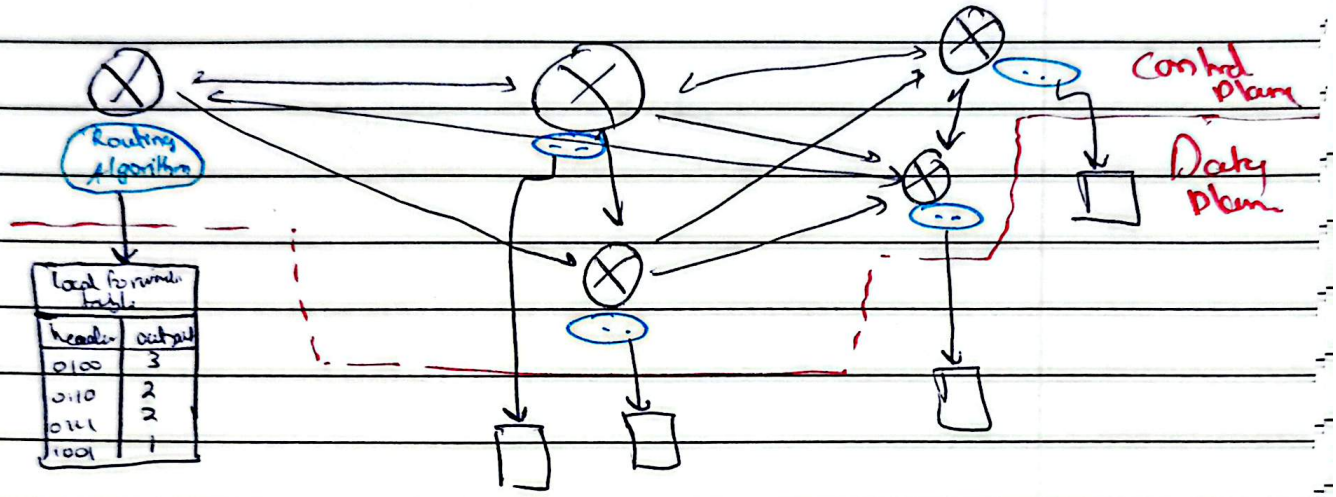
2) Software Defined Networking (SDN):

implemented in remote servers.

1) Traditional approach:



In this every router has a routing Algorithm & each & every router interact with the control plane.

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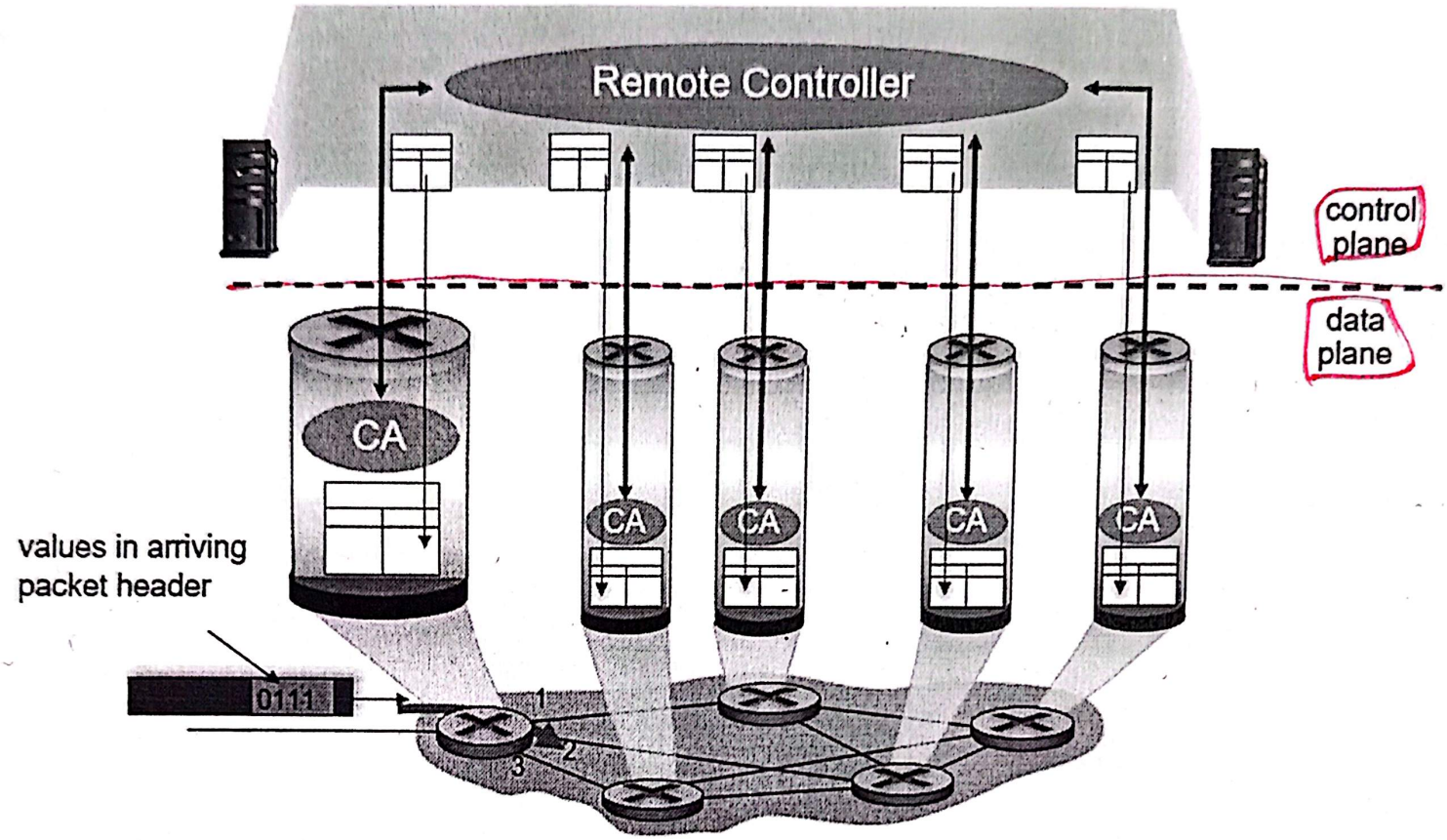
2) SDN approach:



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Network Service Model:

What service model should be used for transporting datagrams & what do they offer?

e.g - consider a service for a single datagram:
- we can guarantee delivery & very low delay:

- consider services for a flow of datagram:
- we can guarantee in-order delivery & minimum bandwidth to flow.

Is it possible to get best of everything? e.g bandwidth, loss, order, ~~cost~~ timing.

For this we have many models.

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1) Best effort model:

No guarantee on:
- → sucessful datagram delivery to destination
- → timing or order of delivery.
- → bandwidth available to end-end flow.

But the best-effort model has huge success:
- → simplicity of mechanism allows internet to widely deployed.
- → sufficient bandwidth allows good performance of realtime apps easily.
- → congestion control.

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Some extra detail:

Before Internet IP model we had ATM model (Asynchronous Transfer Model).
ATM used to be connection-oriented unlike IP being connectionless.
AND ATM didn't use packet switching & instead divided data into fixed 53 byte sized cells.

ATM guaranteed Bandwidth, no loss of cell/packet, & best Quality of Service.
But even though it was so good it could not scale.

2) Constant bit rate.
→ Its architecture is ATM.

3) Available bit rate

4) Intserv Guaranteed

5) Diffserv

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Network Architecture	Service Model	Quality of Service (QoS) Guarantees ?			
		Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no
ATM	Constant Bit Rate	Constant rate	yes	yes	yes
ATM	Available Bit Rate	Guaranteed min	no	yes	no
Internet	Intserv Guaranteed (RFC 1633)	yes	yes	yes	yes
Internet	Diffserv (RFC 2475)	possible	possibly	possibly	no

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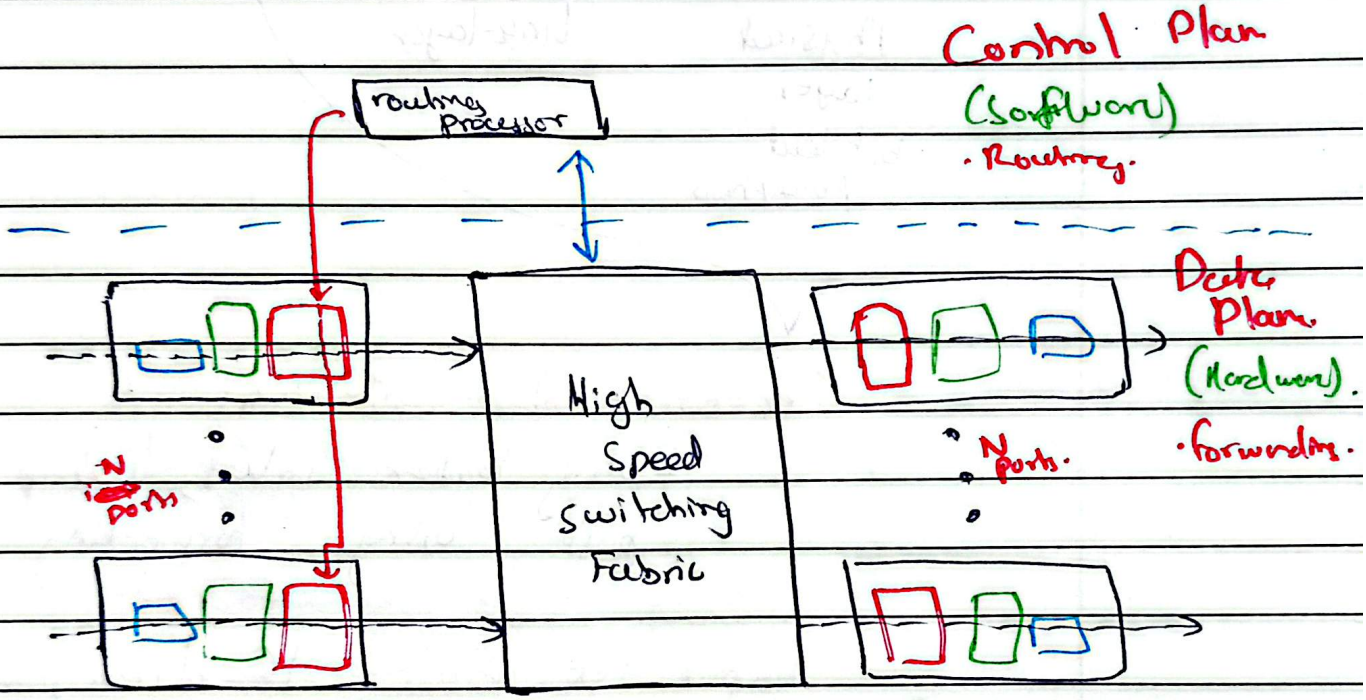
Network layer
Data Plane: Inside a Router

Now we will understand Dataplane & Routers especially & see what is actually inside Routers that allows forwarding.

Router Architecture.



- Like we have said before too, the Data Plane ~~ha~~ has forwarding which is done by hardware that operates in nanoseconds time frame

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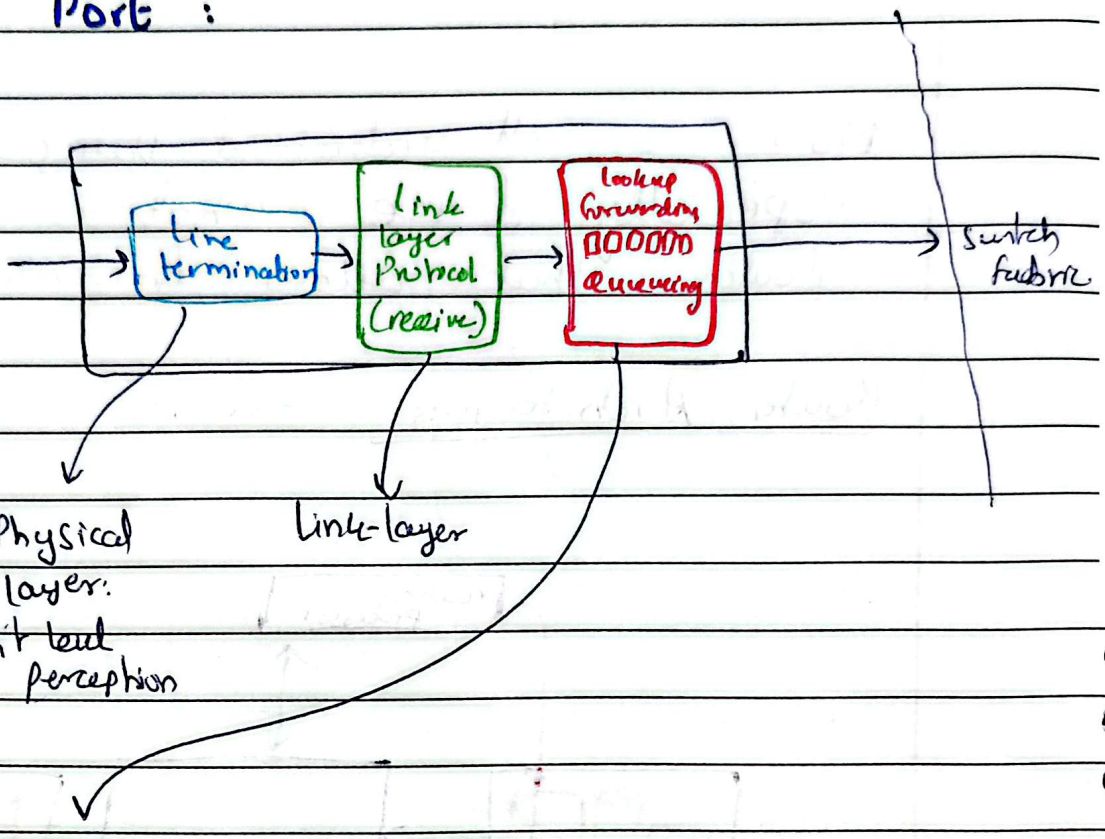
Input Port :



Decentralized switching:

- Using header values, lookup output port using forwarding table

Input Port Queuing: If datagrams arrive faster than forwarding rate into switch fabric.

Destination-based-forwarding:

forward using destination IP address

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generalized forwarding: Forward based on any set of header field values.

we will only learn Destination based forwarding.

Destination - Based forwarding

In this we see the destination Address bits & choose ranges for a particular output.

e.g

Destination Address Range	Link interface
11001000 00010111 00010000 00000000 to 11001000 00010111 00011000 11111111	0
11001000 00010111 00011000 00000000 to 11001000 00010111 00011000 11111111	1
Other wise	2

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But a small problem comes:

what if ranges don't divide up this easily.

e.g

64000 - 112000	0
70000 - 70100	1

} - overlapping ranges.

To prevent overlap we can divide the first range into 2 but that would mean a very big forwarding table.

So to solve ~~this~~ it we keep this configuration but use longest prefix matching.

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Longest Prefix matching:
When looking for forwarding table entry given destination address, use longest adress prefix that matches destination address.

e.g.

Destination address Range	Link interface
11001000 00010111 00010*** ********	0
11001000 00010111 00011000 ********	1
11001000 00010111 00011*** ********	2
Otherwise	3

now

e.g

11001000 00010111 00010110 10100001

matches interface 0. ✓ chosen.

11001000 00010111 00011000 10101010

interface 2.

Interface 1, so it is chosen.

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11001000 00010111 00011001 00000006

Interface 1 matches till here but after this it does NOT match so — interface 1 cannot be taken.
It should be an exact match except for "*".

So interface 2.

Longest Prefix matching is used & performed using TCAM, (Ternary content addressable memory).
these retrieve address in one clock cycle regardless of table size.

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Input port: Important:

bits ⟶ frames ⟶ datagrams ⟶ switching fabric.
(physical layer) (link layer) (network layer)

It converts the bits to datagrams / Packets so we can see network headers & send to particular output link (where its again converted to bits).

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*Switching Fabric.

Okay so now we have the packet from the input ports & now we need to redirect it to the particular output port using the info given by input port (it tells where to go using the forwarding table + longest prefix matching).

Switching fabric just need to move the datagrams packets from input to output.

Switching rate: rate at which packets can be transferred from input to output

- Remeber that there are N input ports & all connected to a switching fabric so best rate ideally is N times the line rate. , NR.

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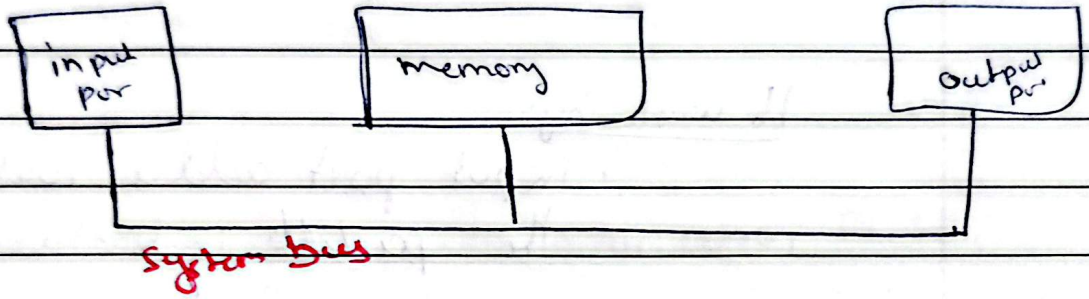
3 major types of Switching fabrics:

1) Switching via memory:



In start, routers were like computers, an incoming packet from input port would trigger an interrupt to CPU. The packet will then be moved to main memory using bus. Then the routing processor looks up output port & copies packet from main memory to output port.

Cons:

- Only one bus so no parallel work
- every packet travels on bus twice so Bus bandwidth/2.

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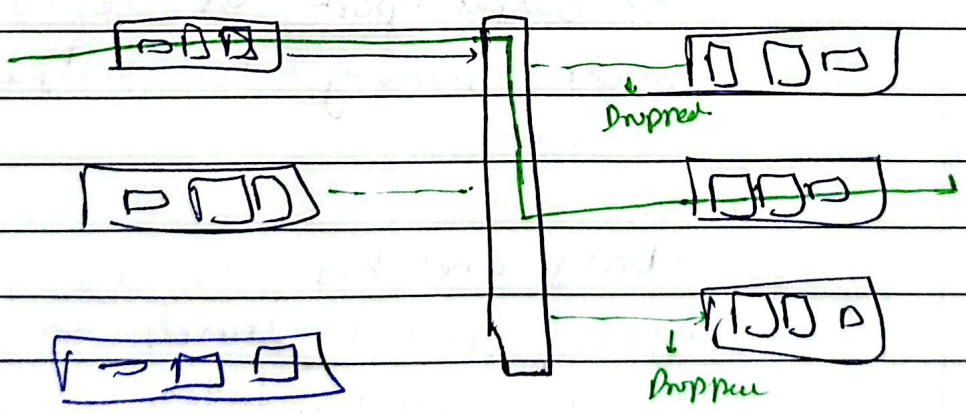
2) Switching via Bus:

Input ports transfer packets directly to output ports over a shared bus without intervention of CPU.

It work by:

- input port adds a routing tag to the packets indicating output port & transmits into the bus.
- All outputs receive it but only the port matching tag keeps it other drop it.
- The routing tag is then stripped off



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Con:

Because only one bus link is shared only one packet data time can traverse.

Router switching speed is capped by bus speed.

3) Switching via Interconnection Network.

To overcome single bus limitation, modern routers use interconnected network such as a cross bar switch.

It works:

- A cross bar consisting of N inputs buses intersecting with N output buses in a 2D grid matrix.
- At each intersection, a switching controller can open or close an electronic switch.

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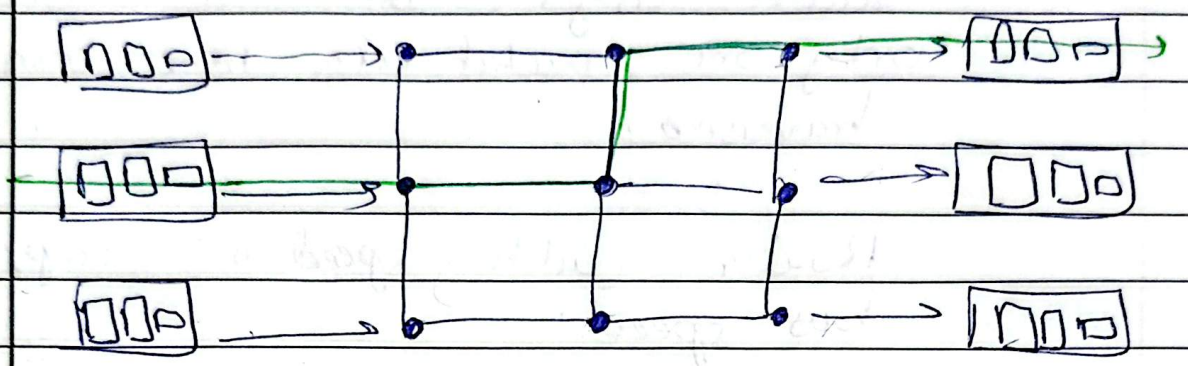

Typically the used to be 3×3 matrix but then they also became too less so it became an 8×8 matrix.

The size of matrix depends upon input Ports N & N output Ports.

But as N increased, the matrix is N^2 so it separated, and it wasn't possible to handle such a big cross bar.

So instead Multiple switching plans started to be used.

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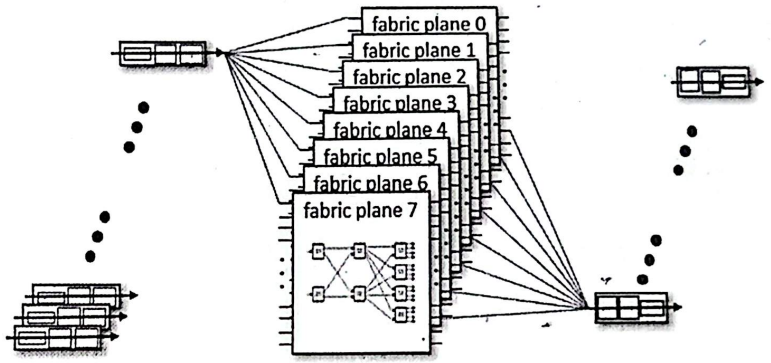
Like multiple crossbars on top of each other.

- scaling, using multiple switching "planes" in parallel:
  - speedup, scaleup via parallelism

- Cisco CRS router:
  - basic unit: 8 switching planes
  - each plane: 3-stage interconnection network
  - up to 100's Tbps switching capacity



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Input port Queuing:

This is a problem that happens when

$N * R_{line} < R_{switch}$.

R_{line} = input link transmission speed.

R_{switch} = switch rate limit

so if our switching rate is lower than line transmission rate then Queuing starts to happen.

Another reason can be: multiple packets across different input ports need to go to same output port, a bottleneck forms.

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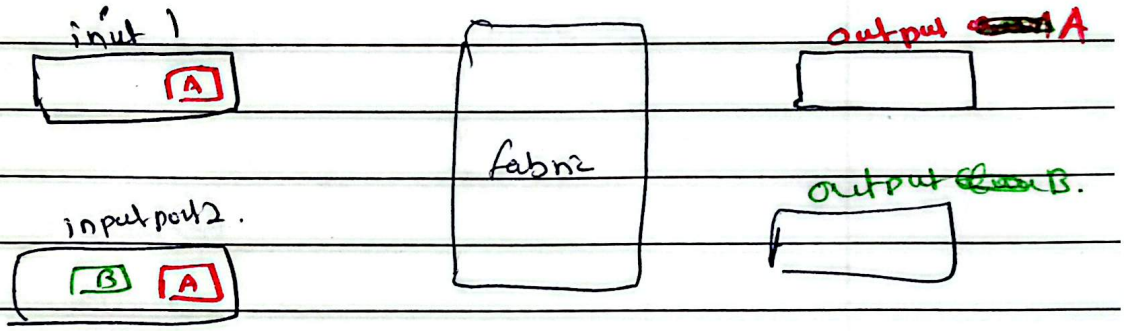
Head-of-line Blocking:

A queued packet at the front of an input queue prevents packets behind it from being forwarded even if the destination output ports for those trailing packets are completely free & idle.

consider scenario



- now both input ports get a packet A at same time. but consider input port 1's is chosen first
Now even though our Output B is free & the Packet B in input Port 2 is blocked bcz of packet A in front of it.

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- Because of HOL input queues build up rapidly even at moderate traffic loads.

- If queue becomes full, packet can be loss.

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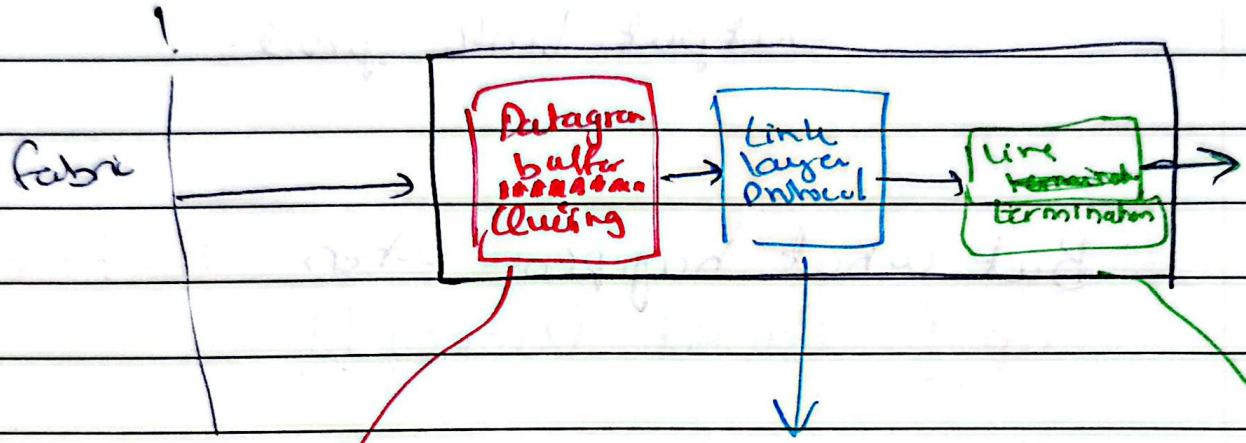
Output port:

Output port is just like input port but opposite



Stores packets delivered by switching fabric if they arrive faster than transmission rate of output link

Encapsulates IP datagram to link-layer frames.

Convert frames bits into physical signals.

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Output Port Queuing:

Buffering: This happens when datagrams arrive faster from fabric than output link speed.

But what happens when **Buffer fills up?** **Packet Drops!!!**

But how to choose which Packet to Drop.

It follows a **Drop policy.**

Buffer management

- e.g. If buffer is full drop the new packet. (Tail drop)
- Instead of waiting for buffer to fill completely, track queue length & begin dropping packets (priority wise). (Priority Drop). or just mark them to signal congestion to end devices.

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How to decide the size of Buffer of output port?

Traditional way:

$$B = RTT \cdot C$$

B = buffer size

RTT = Typical Round trip time

C = Link Capacity (bandwidth) of router's output interface

Modern way:

$$B = \frac{RTT \cdot C}{\sqrt{N}}$$

N = number of unique independent flows / pkt. packets currently flowing through the router.

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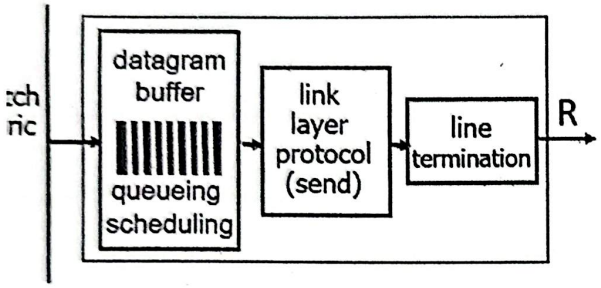
• Too much buffering can increase delays.

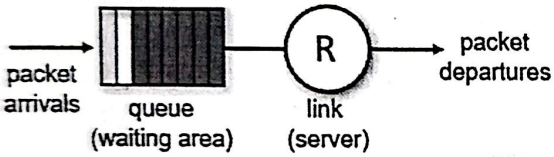
Buffer Management



buffer management:

- drop: which packet to add, drop when buffers are full
  - tail drop: drop arriving packet
  - priority: drop/remove on priority basis
- marking: which packets to mark to signal congestion (ECN, RED)

Abstraction: queue



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Packet Scheduling Policies:

Now for the output buffer, it has to decide which packet to send next.

For this we have multiple policies:

- FCFS (First come, First served).
- Priority
- Round Robin
- weighted fair queuing.

Packet Scheduling: FCFS

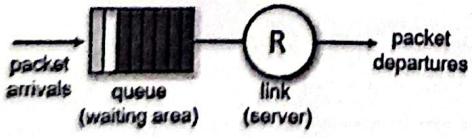
packet scheduling: deciding which packet to send next on link

- first come, first served
- priority
- round robin
- weighted fair queueing

FCFS: packets transmitted in order of arrival to output port

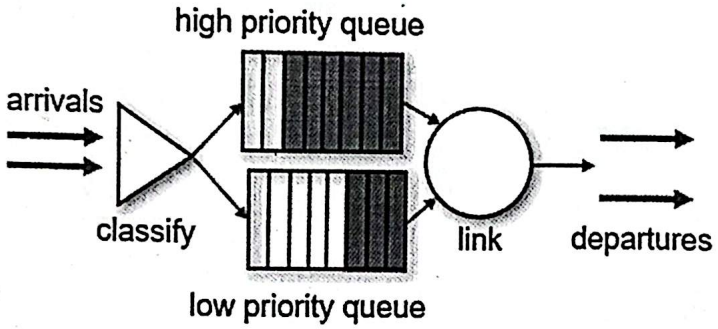
- also known as: First-in-first-out (FIFO)
- real world examples?

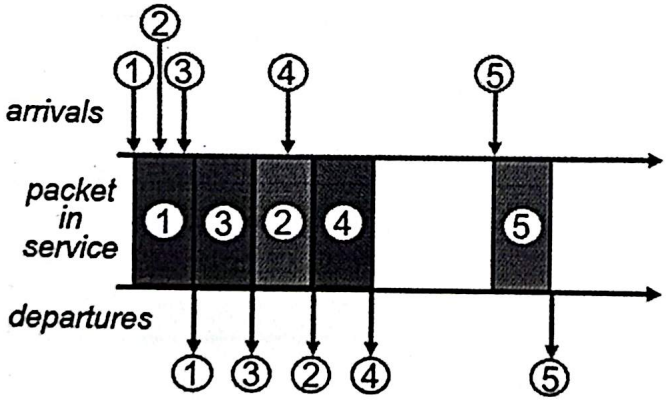
Abstraction: queue



2) Priority:

- Arriving traffic is classified & queued by class.
- Any header field can be used for classification.
- FCFS within a priority class.

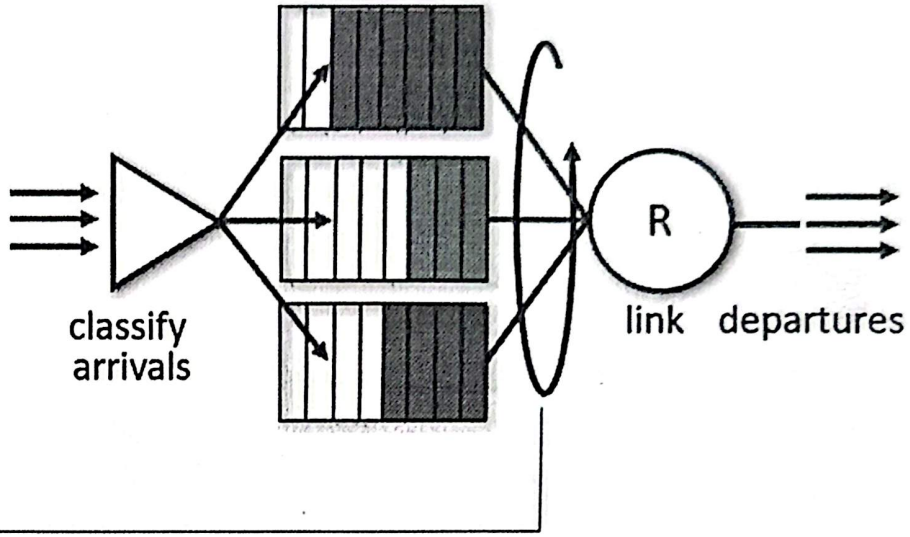




3) Round Robin

Round Robin (RR) scheduling:

- arriving traffic classified, queued by class
  - any header fields can be used for classification
- server cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn



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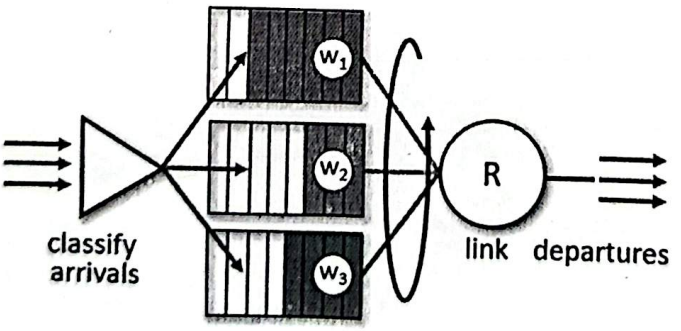
4) Weighted Fair queueing:

This is advanced version of Round Robin.

- Each class i is assigned a weight.
- During any time where all queues have packets class i is guaranteed to receive a fraction of total link bandwidth equal to:

$$\frac{w_i}{\sum w_i}$$

If link speed is R, class i is guaranteed throughput: $R * \frac{w_i}{\sum w_i}$



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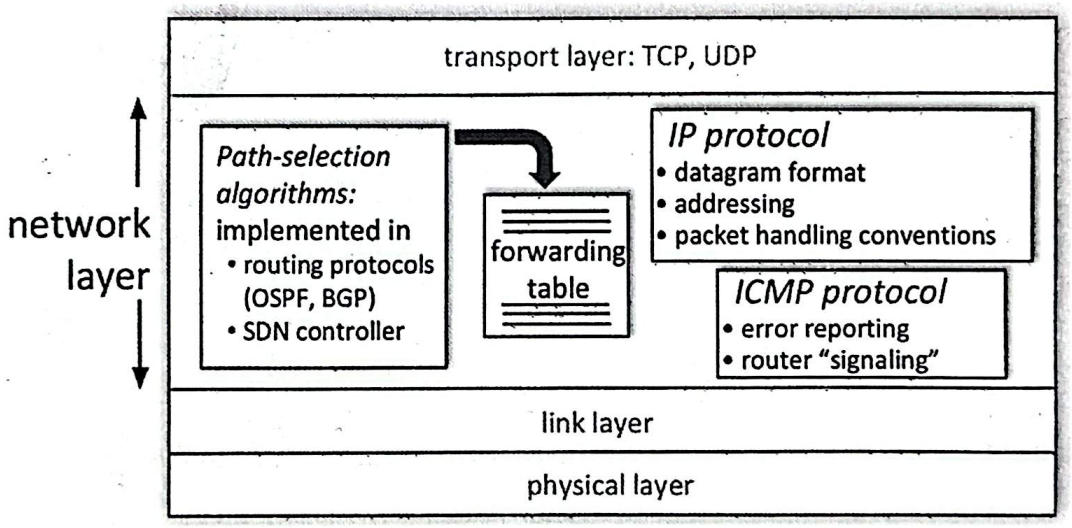
Network Layer

Data Plane: IP: Internet Protocol.

Now talking about the network layer Protocol.

The transport layer had mainly TCP & UDP.
Likewise Network layer also has protocols.

All router network functions:



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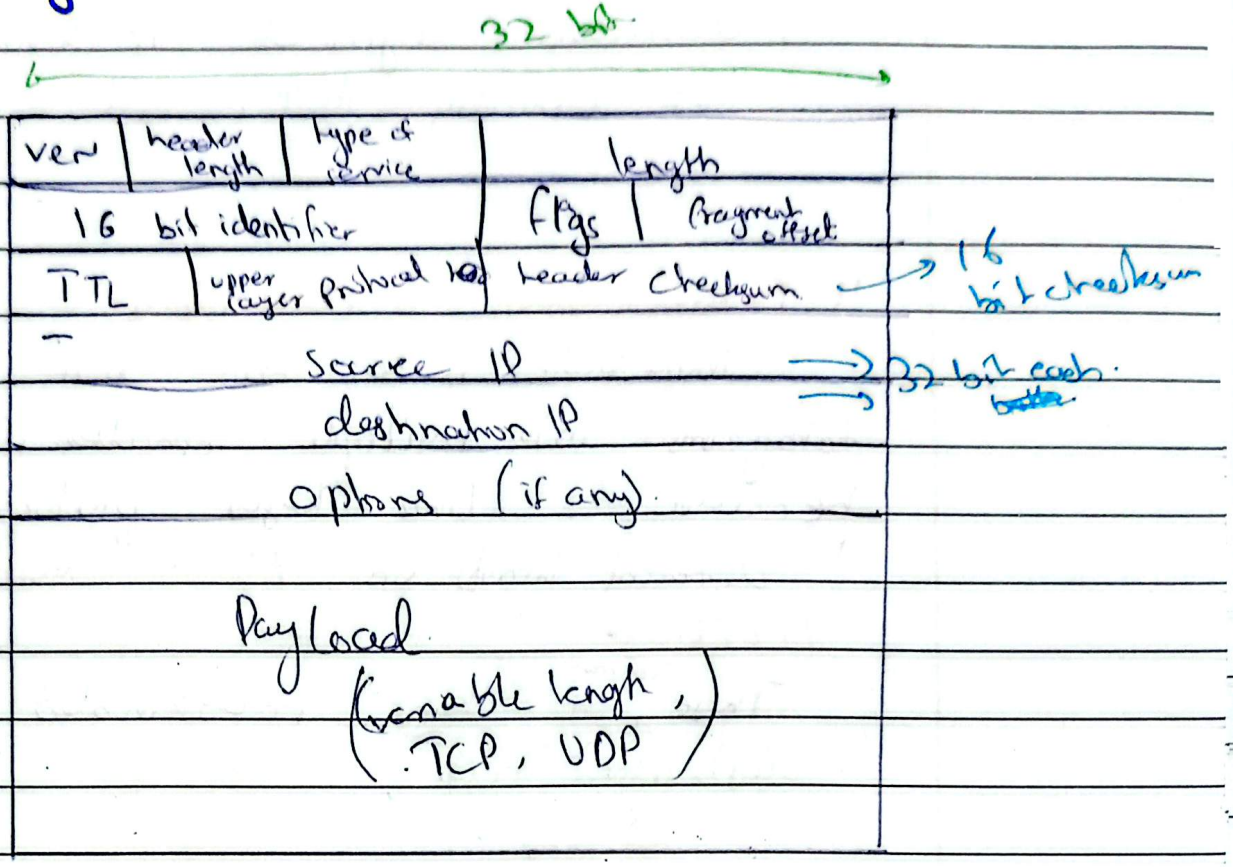
IP Datagram format:



Ver: IP protocol version: e.g IPv4 or IPv6

header length: in bytes.

Type of service: diffserv or ECN

TTL: remaining Max hops. (decrement by 1 at every router)

upper layer Protocol: e.g TCP or UDP

Flgs/16 bit identifier/fragment: used for IP fragmentation & assembly.

options: extra things like timestamp.

IP header is also 20 bytes minimum

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- Maximum length of IP datagram = 64KB
  but typically they are only 1500 B or less.

IP Fragmentation & reassembly:

Immediate routers can split large IP datagrams into multiple smaller IP datagrams because link-layer protocols have different limits on Max transmission time (MTU).

These fragments are then reassembled at the destination host.

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IP addressing: (IPv4)

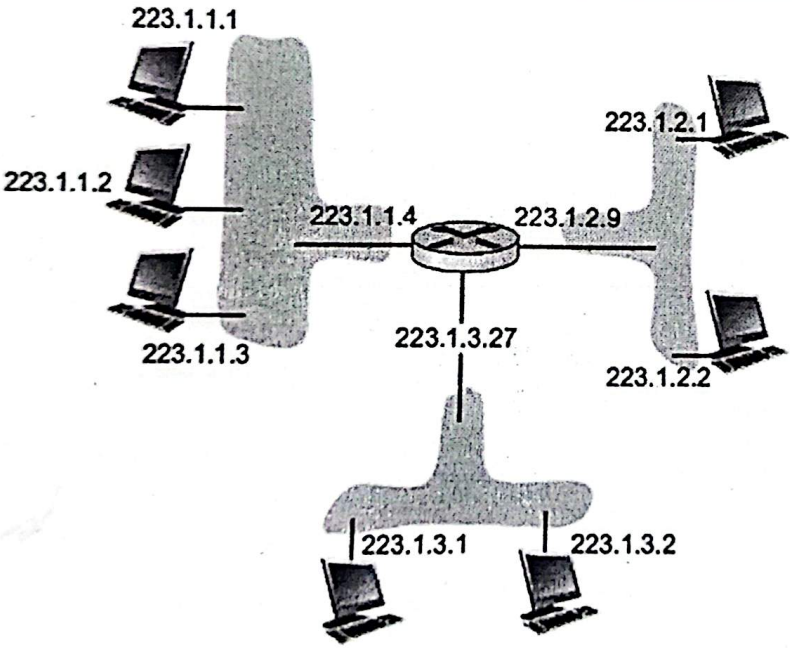
An IPv4 is a 32 bit binary number written by seperating octets (8 bits) by dots:

e.g.

192 . 168 . 1 . 1

8 bits 8 bits 8 bits 8 bits.

- These IP addresses are associated with each host and router interface



dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

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what is an interface?

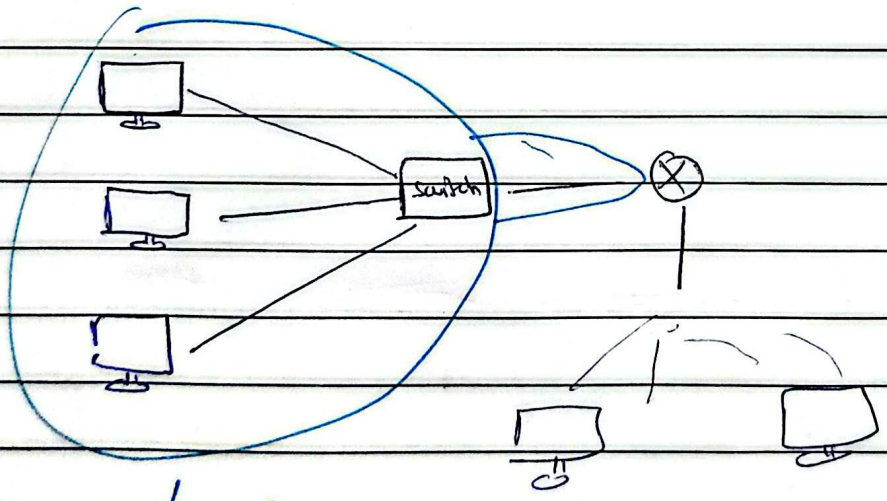
connection between host/router & physical link.

- Host typically has one or interfaces.
- Routers have multiple interfaces.

Subnets?

A subnet are a device interfaces that can physically reach each other without passing through an intervening router.

immediate routers are not intervening routers as well



a single subnet

3 PC's can talk without intervening router

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IP address structure:

There are 2 parts to a IP address:

- Higher order bits : Subnet Part / Network bits
- lower order bits: Host bits.

e.g:

11111110 . 10101010 . 1111000 . 11000001

Network bit | Host bits

These amount of bits can change is both, I just made an example.

- PC's in same subnet will have same subnet part / Network bits but different host bits.

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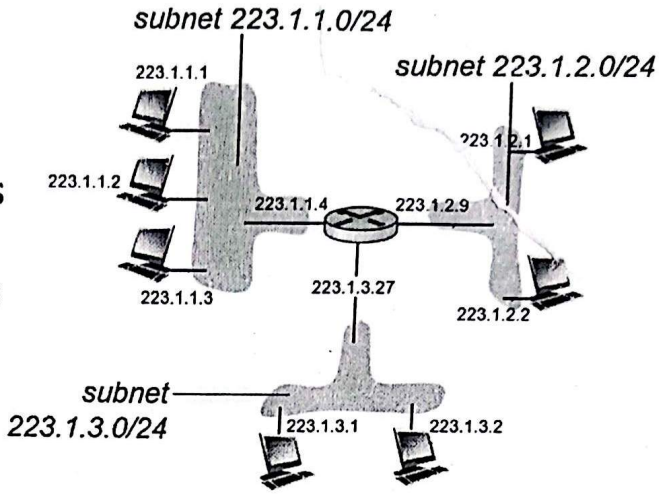
Recipe for defining subnets:

- detach each interface from its host or router, creating "islands" of isolated networks
- each isolated network is called a *subnet*



IP addressing : ~~CIDR~~ Classful addressing:

Before CIDR, the IP addresses were divided into rigid classes (A, B & C) which wasted huge block of addresses & then this led to adoption of CIDR.

~~CIDR: classless~~ ~~Inter Domain Routing~~.

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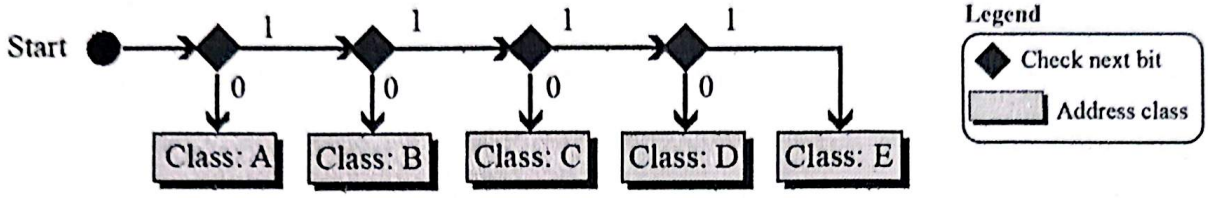
Classful Addressing (The old way of IP addresses)



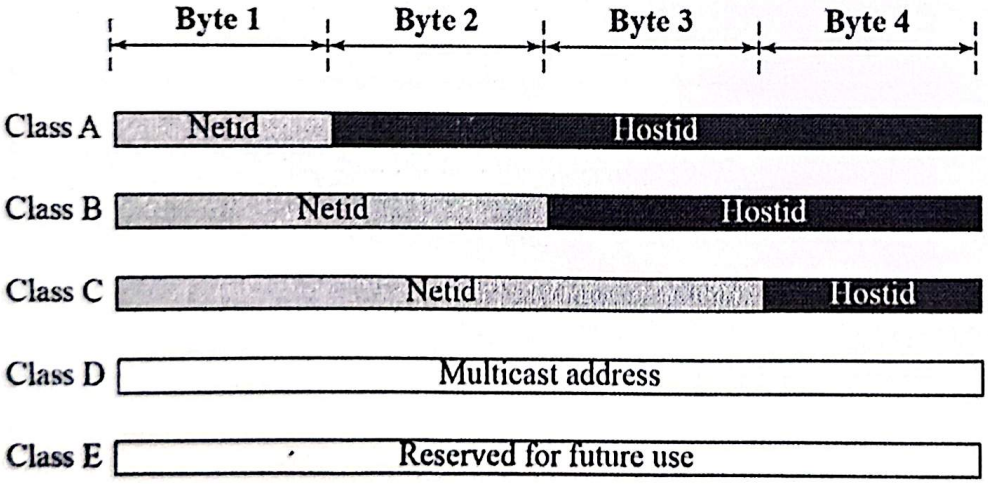
	Octet 1	Octet 2	Octet 3	Octet 4
Class A	0.....			
Class B	10.....			
Class C	110.....			
Class D	1110....			
Class E	1111....			

Binary notation

	Byte 1	Byte 2	Byte 3	Byte 4
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

Dotted-decimal notation

- **IP address in classes A, B, and C is divided into *Netid* and *Hostid*.**
- **These parts are of varying lengths, depending on the class of the address.**



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These wasted a lot of addresses because the net id & host id boundaries were fixed to full byte increments (8, 16, 24 bits).

consider an organisation with 2000 hosts.

class C has ~~host~~ max 254 host & class B has 65534 hosts
This means about 63000 host addresses are wasted.

This all lead to CIDR.

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IP addressing: CIDR:

Classless inter Domain Routing.

IMP:

From all the host addresses available we need to consider network address & broadcast address as well.

network addrs (all Host bits 0)

broadcast addrs (all Host bits 1).

so always consider if we need 200 Hosts, we actually need 200+2 = 202 addresses in total.

Coming back to CIDR

In CIDR, an IP address is written in format

a.b.c.d/x.

a,b,c,d = octets.

/x = prefix length (number of network bits)

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now in this we can assign any number of bits to the network & to host allowing us to save a lot of addresses.

example

192 · 168 · 1 · 130 /25

means

25 = network bit

32 - 25 = 7 Host bits.

11000000 · 10101000 · 00000001 · 10000010

Subnet part

Host part.

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Subnet mask:

A subnet mask is basically all
network bits = 1
&
Host bits = 0.

e.g for /25:

11111111 . 11111111 . 11111111 . 10000000

Subnet Mask = 255 . 255 . 255 . 128

Finding network address:

This is when all
Host bits are 0
& network bits are unchanged.

e.g:
for 192.168.1.130 /25

network address : 192.168.1.128

kept network bits same.

changed all Host bits to 0.

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How to find Total number of Hosts allowed:

Total IPaddresses = $2^{(32-x)}$

Total Usable IPaddress = $2^{(32-x)} - 2$.

(1 network address & 1 broadcast address.)

x = prefix length

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FLSM & VLSM:

FLSM.

FLSM: Fixed length subnet masking.

In FLSM every subnet created from a network block uses exact same subnet mask, so all subnets get exact same amount of IP addresses.

- But this leads to massive address waste when ~~depart~~ ~~ments~~ departments have vastly different host requirements.

Question:

consider an organization with network IP: 192.168.10.0 /24

Now ~~need to~~ organisation has 4 departments:

A needs 64 addresses Total
B need 20 " "
C " 35 " "
D needs 5 address in total.

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Because we are using FLSM, we need to see largest one:

A – 64. (total addresses).

(total addresses means network & broadcast included).

so $64 = 2^6$

so 6 host bits are needed.

so /26.

Total amount of subnets that can be made:

8 – 6 = 2 network bits extra

8 → Total Host bits

6 → needed.

Total subnets: $2^2 = 4$.

so all 4 department can have their own subnet.

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~~A gets 192.168~~

$2^6 = 64$ bits

each subnet get 64 bits.

A gets : 192.168.10.0 — 192.168.10.63

B gets : 192.168.10.64 — 192.168.10.127

C gets : 192.168.10.128 — 192.168.10.191

D gets : 192.168.10.192 — 192.168.10.255

But because we divided equally.
so many addresses got wasted
e.g D got 64 but needed only 5 so 59 wasted.

So to fix this we have VLSM.

All steps in FLSM:

- see max addresses department.
- Find host bits needed.
- Find subnets that can be made
  (2^s) s = total host bits − required host bits
- Check if departments ≤ subnets
  (Possible or not possible).
- give addresses range

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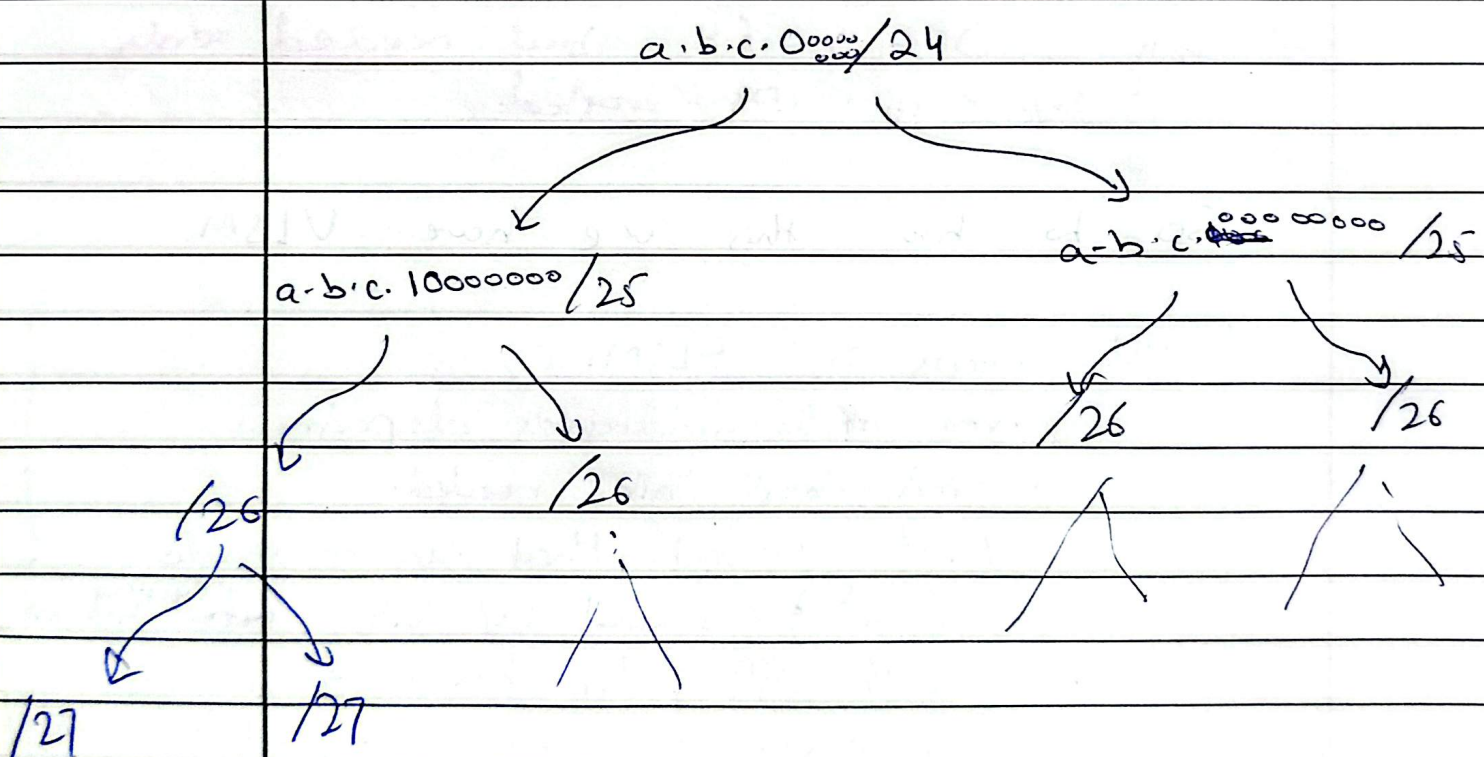
VLSM: (Variable length subnet masking)

This allows an organization to use different subnet masks for different subnets.

For the network provided we keep on breaking it increase one bit of network at a time.

e.g

If network is /24 for organization



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So this is how we actually do in VLSM, we keep on dividing & consider it a network need, /25 we give it that one branch, but other branch keep growing till all requirements are fullfilled.

But still even in this, we should see possibility.

e.g if network address is /24,

Then only 2*(/25) are possible

or

1*(/25) and 2*(/24) are possible

or

4*(/24) are possible

1/8 soon.

So we need to check this possibilities as well.

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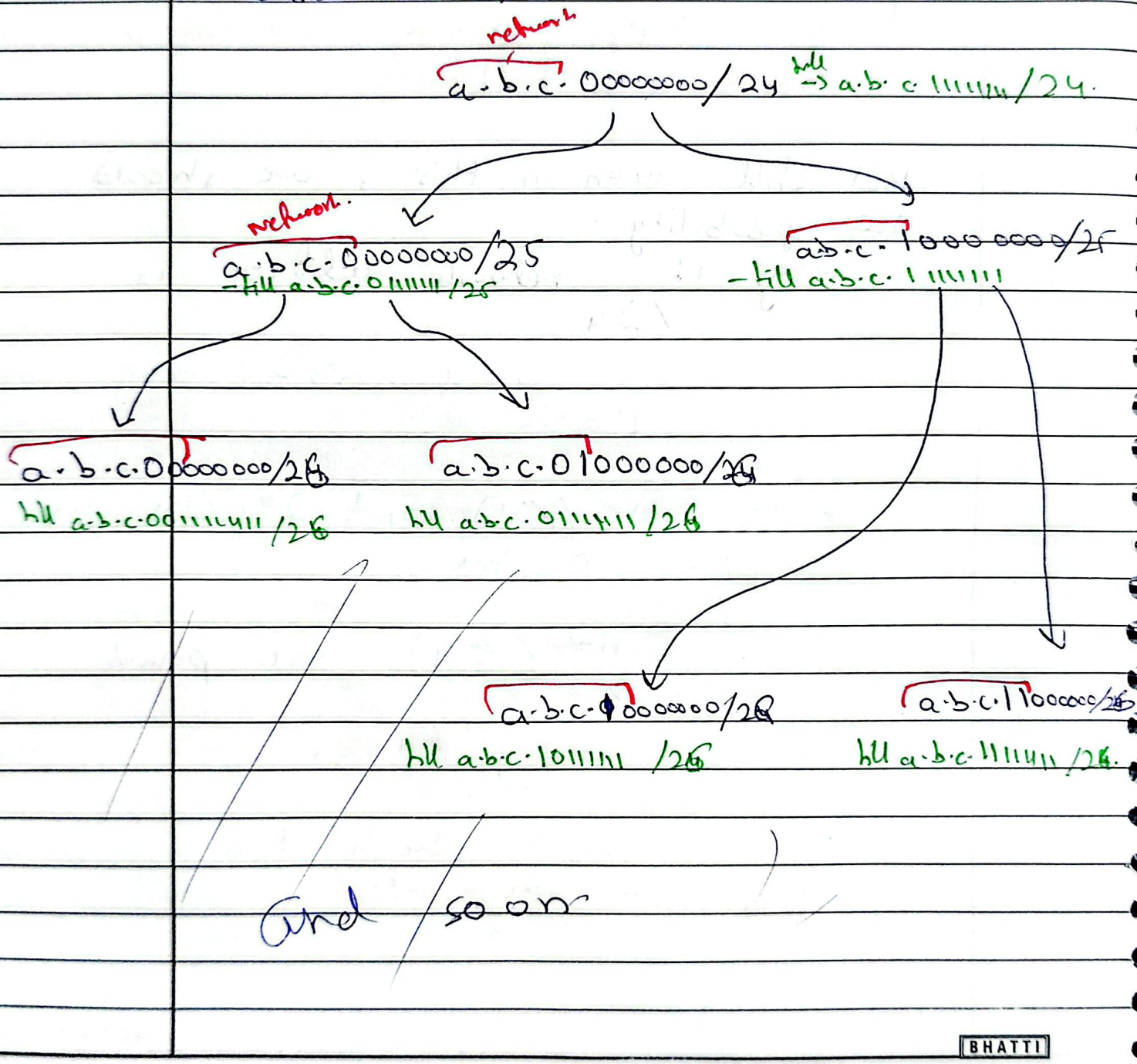
Tree method of solving VLSM:

This is the tree we made behind consider for /24.



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if 1 bit decimal:

a.b.c.0/24 - a.b.c.255/24

a.b.c.0 - a.b.c.127 /25

a.b.c.128 - a.b.255 /25

a.b.c.0 - a.b.c.63

a.b.c.64 - a.b.c.127.

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Question: Solving in VLSMs

* Remember one more thing:
assign lower addresses first (not a rule, but a better practice).
What I mean is if we have 2 brandes 0-127 & 127-255 assign 0-127 first & then check if we need to devide 127-255 or assign it as well.

Question

An organization has IP address:
100.50.50.50/24

Requirement by departments

A → 13 Hosts.
B → 63 Hosts.
C → 2 Hosts.
D → 7 Hosts
E → 1 Host.
F → 5 Hosts.
G → 11 Hosts.

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Step 1: (as the question says required Hosts & not total addresses so First add the 2 extra hosts' needed).

A → 15 addresses needed.
B → 65 " "
C → 4 " "
D → 9 " "
E → 3 " "
F → 7 " "
G → 13 " "

Step 2: (Optional) (arrange in descending order: Helps).

B → 65
A → 15
G → 13
D → 9
F → 7
C → 4
E → 3

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Step 3: (write what power of 2 or what subnet (/x) can each ~~add~~ ~~Hosts~~ department com in).

needed bits.

B → 65 → 2^7 so 128 so /25
A → 15 → 2^4 so 16 /28
G → 13 → 2^4 so 16 /28
D → 9 → 2^4 so 16 /28
F → 7 → 2^3 so 8 /29
C → 4 → 2^3 so 8 /30
E → 3 → 2^2 so 8 /30.

- even though all addresses are wasted we save alot bro.

* Step 1 to 3 can be done in a single step.

* also after this you can check if the subnetting is possible or not.

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Step 4: Make tree:

- Find network address of organisation:
  150.50.50.0 /24
- Make the tree but whenever we find a branch whose subnet matches one of the department. Stop expansion & assign it to the department.

- First the lower addresses branch is assigned is assigning is needed. (optional but better this way).

Tree on next page.

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100.50.50.0/24
-100.50.50.255/24

100.50.50.0/25
100.50.50.127/25
assigned to department B.

B had /25
so it is assigned
this whole block.

• only B in /25
so that
if occupied
• keep branching
the other branch.

100.50.50.128/25
100.50.50.255/25

no Department in /26.

100.50.50.128/26
-100.50.50.191/26

no branch in /27

100.50.50.191/26
-100.50.50.255/26

100.50.50.128/27
-100.50.50.159/27

100.50.50.160/27
-100.50.50.191/27

100.50.50.128/28
-100.50.50.143/28
assigned to A

100.50.50.144/28
-100.50.50.159/28
assigned to G.

100.50.50.160
-100.50.50.175
assigned to D.

100.50.50.176/28
-100.50.50.191/28

no branch /28 so branch.

100.50.50.176/29
-100.50.50.183
assigned to 1st

and keep on going

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We hit all Department assigned ref.

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(marked on tree)

* I left this branch bcz no space & for sake of understanding & question it wasn't needed. It would just be not used.

. Tree would either end if all ~~departments~~ assigned or all branches ended.

If branches ended & not all department assigned then it means that this network something is not possible.

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IP address: How does a host get an IP?

For host to get an IP, we do understand subnetting but it gives a range of usable IP's.
which one does the host get?

2 ways:

1) Hard coded by system admin in config.

2) DHCP (dynamically allocated)

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DHCP: (Plug & play connectivity)
"Dynamic Host configuration Protocol"

In this host dynamically obtains IP addresses from network server when it "joins" the network

Pros:

- can renew its lease on address in use.
- allows reuse of addresses
- support for mobile users who join/leave network.

* It is an applayer protocol

* Operates over UDP using Port 67 (server) & port 68 (client).

* In this there is a server ~~host~~, a DHCP server, colocated in router, serving all subnets to which router is attached.

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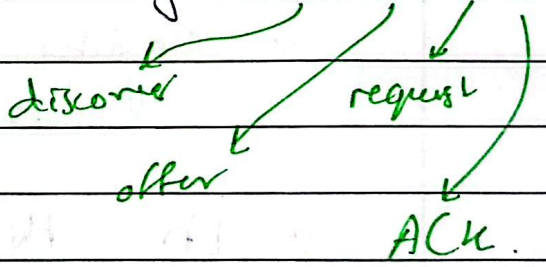
DHCP connection establishing & IP assigning.

4 steps:

- The client broadcasts DHCP discover (D) message (optional).
- The DHCP server responds with DHCP Offer message (O) (optional).
- host requests IP address by a DHCP request message (R)
- Server sends address : DHCP ack message (A).

To revise use acronym: DORA



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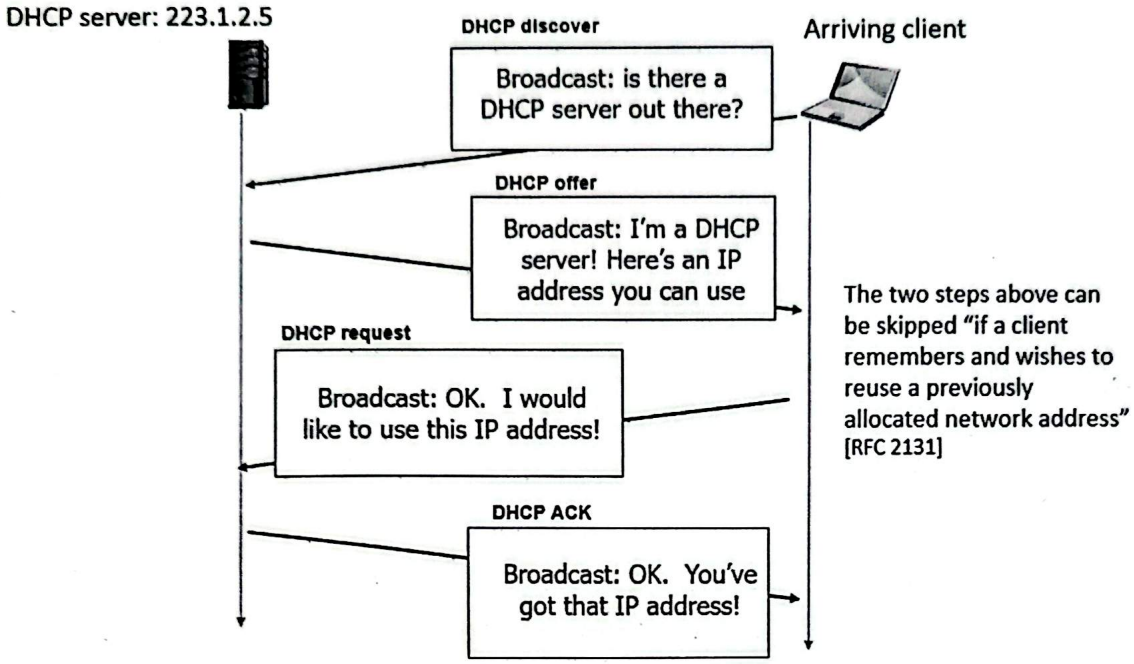

DHCP services:

- It can do ~~not~~ a lot more than returning allocating IP addresses on subnet
- can give address of first hop router
- name & IP of DNS server
- network mask.

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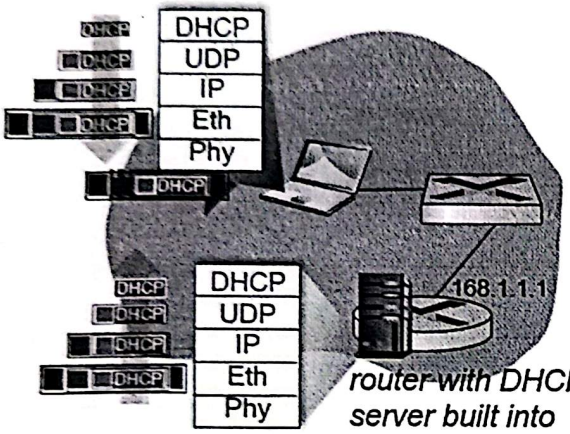
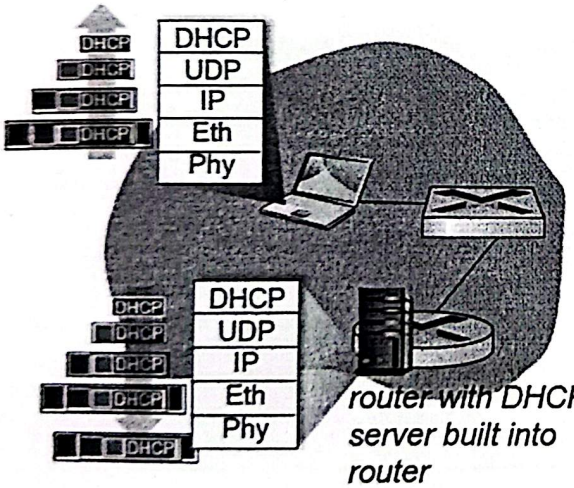
DHCP example



router with DHCP server built into router

- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP



router with DHCP server built into router

- DCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

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IP addressing: How does network get subnet Part of IP address.

We understood how Host get IP using DHCP.

For when does the network bit come & who assigns those?

Each Network / organisation gets allocated portion of its provider ISP's address space.

e.g consider ISP's network

200.23.16.0 /20

now it can give from its address space to organisation using subnetting:

ISP can then allocate out its address space in 8 blocks:

Organization 0	11001000 00010111 00010000	00000000	200.23.16.0/23
Organization 1	11001000 00010111 00010010	00000000	200.23.18.0/23
Organization 2	11001000 00010111 00010100	00000000	200.23.20.0/23
...	.....	....	....
Organization 7	11001000 00010111 00011110	00000000	200.23.30.0/23

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Hierarchial Addressing:

We know that an address is divided into network bits & Host identifier.

Now consider if each router stored every entry/address in its routing table.
Routing table would consum memory in GB's & lookup time would be huge.
This is the problem of flat addressing.

To fix it we have hierarchial addressing where network bit & host come in handy.

In hierarchial addressing: routers only care about the network prefix, ignoring host bits.

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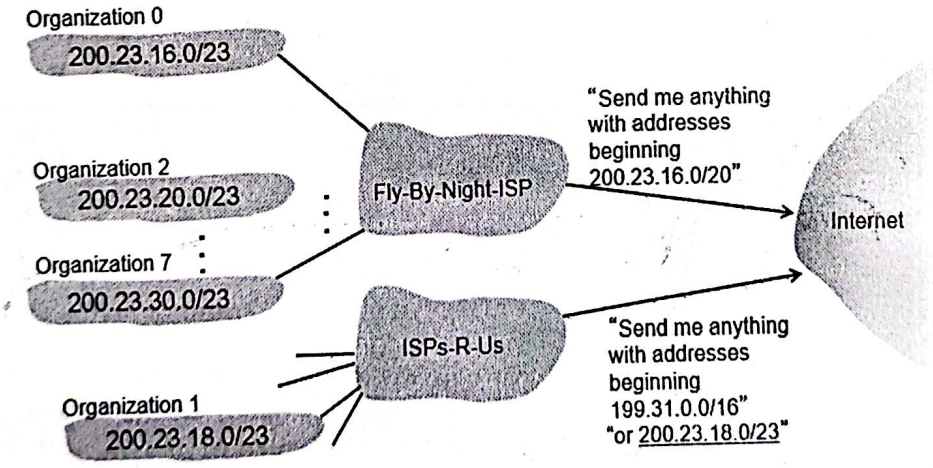
This makes a hierarchy:

- Global internet router: only contain entries pointing to ISP routers.
- ISP internal router: have entries pointing to organization networks.
- Organization router can divide their networks further using VLSM or FLSM

Example:



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- This benefit reached by hierarchical addressing is called : Route aggregation

This means that all organizations networks are included in ISP's network.

So the main global router ~~does~~ only needs the aggregated ISP network stored in it.

IP address: Some extra stuff:

Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers http://www.icann.org/

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , ...) management

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)

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Network address translation (NAT)

NAT is an IP management mechanism that allow an entire private local area network To share a single public ipv4 address. or a small pool of public addresses. when communicating with the outside internet.

Private vs Public addresses:

Private	Public
• Used with LAN or network	• Used on Public Network
• Not recognized over Internet	• Recognized over internet.
• Unique in LAN	• Unique globally.
• Free of Charge	• has Cost

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Why NAT exists & ranges assigned.

To prevent every individual device on earth from requiring a globally unique & public IP address RFC 1918 designated 3 specific blocks for private networks

1) 10.0.0.0/8 – 10.255.255.255/8
2) 172.16.0.0/12
3) 192.168.0.0/16.

NAT & benefits:

All devices in local area only talk through one IPv4 address but there source port number different.

- One IP needed from ISP for all devices.
- can change addresses of host in LAN with keeping global known
- security: as devices in LAN not directly addressable.

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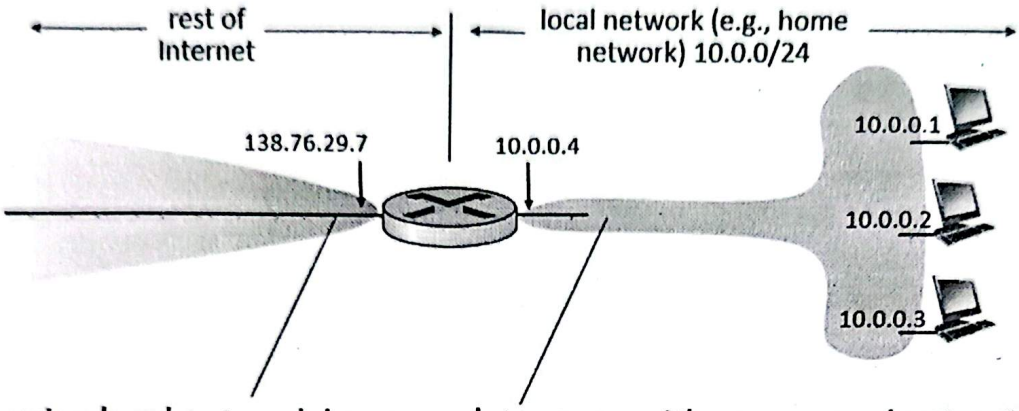
- This translation is done by the NAT router.



all datagrams *leaving* local network have *same* source NAT IP address: 138.76.29.7, but *different* source port numbers

datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

NAT Routers

How it works?

- Replace (source IP, port #) with (NAT IP, new port #) on out going datagram.
- Remember every (source IP, port #) ⟷ (NAT IP, new port #) pair. (It has a table inside to store it all)

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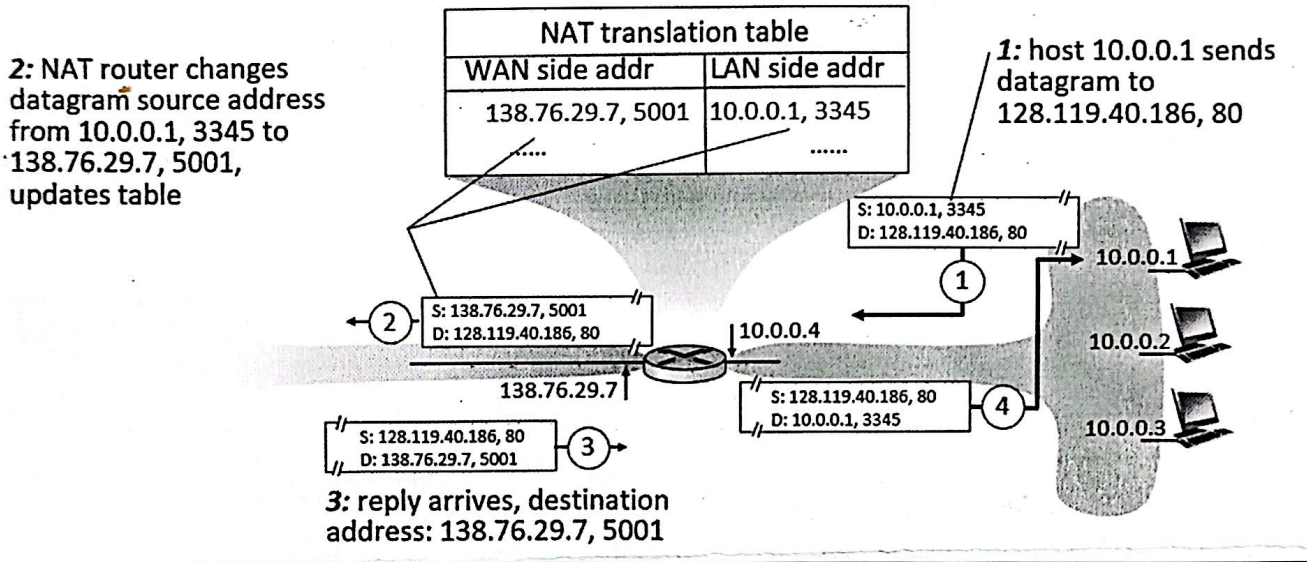
- Replace (Nat IP, new port#) with (Source IP, port#) for incoming datagrams.

- Every WAN router has a NAT translation table inside.



- The server/responder can also be behind NAT.
- Servers & respond back to public IP not private IP.

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NAT controversies.

* NAT also has port # but routers only process upto layer 3. not 2.
* address shortage should not be solved by NAT. but by ipv6
* what if Client wants to connect to server behind NAT
* violates end-end argument (Port # manipulation by network layer)

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